Chapter 6, Problem 37P

Problem

The current through a 12-mH inductor is 4 sin 100*t* A. Find the voltage, and the energy stored at $t = \frac{\pi}{200}$ s.

Step-by-step solution

Step 1 of 3

Write the expression of current flowing through inductor.

$$i(t) = 4\sin(100t)\,\text{A}$$

The inductor value is, $L = 12\,\text{mH}$

Determine the voltage across inductor.

$$v = L\frac{di}{dt}$$

Substitute $12\,\text{mH}$ for L and $4\sin(100t)\,\text{A}$ for $i(t)$.

$$\begin{aligned} v &= \left(12\times10^{-3}\right)\frac{d}{dt}\left[4\sin(100t)\right] \\ &= \left(12\times10^{-3}\right)\left[400\cos(100t)\right] \\ &= 4.8\cos(100t)\,\text{V} \end{aligned}$$

Voltage at $t = \frac{\pi}{200}$ sec is

$$v = 4.8\cos\left(100\times\frac{\pi}{200}\right)$$

$$v = 0\ \text{V}$$

Voltage across inductor at $t = \frac{\pi}{200}$ sec is $\boxed{0\ \text{V}}$

Step 2 of 3

Determine the power in the inductor.

$$p = vi$$

Substitute $4.8\cos(100t)\,\text{V}$ for v and $4\sin(100t)\,\text{A}$ for $i(t)$.

$$\begin{aligned} p &= \left(4.8\cos(100t)\right)\left(4\sin(100t)\right) \\ &= 19.2\cos(100t)\sin(100t) \\ &= \frac{19.2}{2}\left[2\cos(100t)\sin(100t)\right] \\ &= 9.6\left[\sin 2(100t)\right] \qquad \left[\text{Since } 2\cos(\theta)\sin(\theta) = \sin(2\theta)\right] \end{aligned}$$

$$p = 9.6\sin(200t)$$

Step 3 of 3

Determine the energy stored in the inductor.

$$\begin{aligned}
w &= \int_0^t p\,dt \\
&= \int_0^{\frac{\pi}{200}} 9.6\sin(200t)\,dt \\
&= \frac{9.6}{200}\left[-\cos(200t)\right]_0^{\frac{\pi}{200}} \\
&= -\frac{9.6}{200}\left[\cos(\pi)-\cos(0)\right] \\
&= -\frac{9.6}{200}\left[-1-1\right] \\
&= 0.096 \\
&= 96\,\text{mJ}
\end{aligned}$$

Therefore, the energy stored in the inductor at $t=\frac{\pi}{200}\,\text{s}$ is $\boxed{96\text{ mJ}}$.